Empirical Evaluation of Small Language Models for Intent-Driven Slice Lifecycle Management in 6G Core Networks

Teerapong Somsri, [Co-authors TBD]

Rajabhat Ratchanakarin University (RBRU)

Abstract

The deployment of intent-based networking (IBN) in 6G core networks requires lightweight, efficient language models capable of translating natural language intents into network orchestration commands. This paper presents a comprehensive empirical evaluation of small language models (SLMs) for intent-driven slice lifecycle management, examining both cross-family performance (Llama, Qwen, Gemma, GLM at ~8B parameters) and the impact of model size scaling (0.5B-7B) with quantization (Q2_K, Q4_K_M, Q8_0). Using a dataset of 200 intents spanning provisioning, scaling, and conflict resolution domains, we evaluate models across seven dimensions: format validity, action accuracy, exact parameter match, inference latency, throughput, memory consumption, and energy efficiency. Our results reveal that Qwen2.5-7B achieves the highest overall accuracy (29.8% exact match, 43.5% action accuracy) while Qwen2.5-3B-Q4_K_M offers the best cost-effectiveness ratio. We also identify diminishing returns beyond 3B parameters and find that quantization degradation (Q8_0 to Q2_K) impacts accuracy by only 1-3 percentage points while reducing energy consumption by up to 40%. These findings provide practical guidelines for deploying SLMs in resource-constrained 6G edge environments.

Keywords: Small Language Models, 6G Networks, Intent-Based Networking, Network Slicing, Model Quantization, Edge AI, Resource Efficiency

1. Introduction

Sixth-generation (6G) networks are expected to support diverse and demanding use cases including enhanced mobile broadband (eMBB), ultra-reliable low-latency communications (URLLC), and massive machine-type communications (mMTC). Network slicing — the ability to create multiple virtualized and independent logical networks on a shared physical infrastructure — is a cornerstone technology for meeting these heterogeneous requirements [1,2].

Intent-based networking (IBN) has emerged as a promising paradigm that allows network operators to express high-level business goals in natural language, which are then automatically translated into low-level configurations [3]. Large language models (LLMs) have demonstrated remarkable capabilities in understanding and generating structured outputs from natural language, making them natural candidates for the intent translation task in 6G IBN systems [4,5].

However, deploying LLMs in 6G edge environments presents significant challenges. The computational and memory requirements of models with 7-70 billion parameters are often incompatible with the resource constraints of edge nodes, where power budgets may be as low as 10-50W and memory limited to a few gigabytes [6]. Small language models (SLMs) with 0.5-8B parameters, combined with quantization techniques (2-8 bits), offer a potential solution — but their performance on domain-specific networking tasks remains underexplored.

This paper addresses four research questions:

- RQ1: How do different SLM architectures (Llama, Qwen, Gemma, GLM) compare in intent translation accuracy?
- RQ2: What is the relationship between model size (0.5B-7B) and intent translation performance?
- RQ3: How does quantization (Q2_K, Q4_K_M, Q8_0) affect the accuracy-efficiency trade-off?
- RQ4: What are the resource consumption characteristics of SLM inference on CPU-only hardware?

We answer these questions through a systematic, reproducible benchmark of 16 model configurations on identical CPU-only hardware. Our contributions include: (1) the first comprehensive comparison of open-weight SLMs for 6G intent translation, (2) a detailed Pareto frontier analysis across model sizes and quantization levels, (3) quantitative evidence of diminishing returns beyond 3B, and (4) practical deployment recommendations for resource-constrained 6G edge infrastructure.

2. Related Work

2.1 Intent-Based Networking in 6G

IBN has been extensively studied in the context of 5G and beyond. ETSI ZSM [7] and 3GPP [8] have standardized frameworks for intent-driven closed-loop automation. Recent works by Mechtri et al. [9] and Abbas et al. [2] have proposed natural language interfaces for network slice management. However, these approaches typically assume cloud-based LLMs, which are impractical for edge deployment.

2.2 Small Language Models for Domain Tasks

The emergence of efficient open-weight models such as Llama 3.1 [10], Qwen2.5 [11], Gemma 2 [12], and GLM-4 [13] has made SLM deployment feasible on consumer hardware. Recent benchmarks have evaluated these models on general NLP tasks, but domain-specific evaluation for networking applications remains scarce.

2.3 Model Quantization for Edge Deployment

The GGUF format [14] with K-quant methods has become the standard for CPU inference. Dettmers et al. [15] demonstrated that 4-bit quantization preserves model quality for most NLP tasks. Our work extends this analysis to the networking domain.

2.4 Resource-Aware Model Selection

Prior work on energy-aware ML has focused on training efficiency [16] or cloud inference cost [17]. For edge deployment, memory bandwidth and CPU utilization are primary bottlenecks [18]. Our work contributes per-intent CPU, memory, and energy measurements across 12 configurations on identical hardware.

3. Methodology

3.1 Dataset

We constructed a dataset of 200 natural language intents spanning three network slice lifecycle domains: Slicing Provisioning (70 intents), Scaling Request (70 intents), and Conflict Resolution (60 intents). Each intent consists of an input_text and a ground_truth JSON object containing expected action, slice type, and domain-specific parameters.

Example Intent: Input: "Create a new eMBB slice for a 4K video streaming event in Bangkok." Ground Truth: {"action": "CREATE", "slice_type": "eMBB", "bandwidth_gbps": 1}

3.2 Models

We evaluate models across two experimental phases. Phase 1 compares four open-weight SLM families at ~8B parameters under Q4_K_M quantization: Llama 3.1-8B, Qwen2.5-7B, Gemma 2-9B (custom Q4_K_M), and GLM-4-9B (custom Q4_K_M). DeepSeek-R1-8B was excluded due to CPU-only incompatibility — its reasoning architecture caused infinite chain-of-thought loops on CPU inference [20].

Phase 2 systematically evaluates Qwen2.5 (selected as the top performer from Phase 1) across four sizes (0.5B, 1.5B, 3B, 7B) and three quantization levels (Q2_K, Q4_K_M, Q8_0), resulting in 12 configurations.

3.3 Metrics & Statistical Framework

We define seven evaluation metrics: Format Validity (valid JSON with required fields), Action Accuracy (correct CREATE/MODIFY/DELETE), Avg Match% (mean percentage of ground-truth key-value pairs correctly predicted), Inference Time (wall-clock seconds), Throughput (tokens/s), Memory Footprint (peak RSS in MB via psutil), and Energy Consumption (estimated as CPU_utilization × TDP_135W × inference_time). For every continuous metric, we report the mean, standard deviation, and 95% confidence interval [$\mu \pm t_{0.025} \cdot \sigma/\sqrt{n}$] across n = 200 intents using the t-distribution (df = 199, t ≈ 1.972).

3.4 Hardware & Software

All experiments run on identical Proxmox VE 8.2 KVM virtual machines with 16 vCPUs (Intel Xeon E5-2690 v4 @ 2.60GHz), 32GB RAM, and 80GB SSD. Software: Ubuntu 24.04, Ollama 0.5.x (llama.cpp backend), Python 3.12, psutil 6.x. Each configuration is evaluated independently with a 2-intent warmup to ensure consistent model loading and KV-cache state.

4. Phase 1: Cross-Family Model Comparison

Table 1 presents the results of the cross-family comparison at ~8B parameters with Q4_K_M quantization.

Model	Format %	Action %	Avg Match %	Time (s)
Llama 3.1 8B	100.0	36.5	26.0	19.11
Qwen2.5 7B	100.0	43.5	29.8	18.68
Gemma 2 9B	98.5	45.5	29.3	39.79
GLM-4 9B	100.0	29.0	24.6	22.88

Table 1: Phase 1 cross-family benchmark results (~8B, Q4_K_M, n=200 intents).

4.1 Key Findings

- Qwen2.5-7B achieves the best overall balance: highest exact match (29.8%) with lowest inference time (18.68s), representing a 14.6% relative improvement over Llama 3.1.
- Gemma 2-9B has the highest action accuracy (45.5%) but at 2.1× the latency of Qwen (39.79s vs 18.68s), making it unsuitable for latency-sensitive edge applications.
- GLM-4-9B significantly underperforms in action accuracy (29.0%), suggesting its instruction-tuning may be less aligned with structured JSON output tasks.
- The 9B models (Gemma, GLM) do not show proportional gains over 7-8B models, indicating that architecture and training quality outweigh raw parameter count for this task.

Based on these results, Qwen2.5 is selected as the reference architecture for Phase 2.

5. Phase 2: Size Scaling & Quantization

5.1 Complete Results with 95% Confidence Intervals

Table 2 presents the complete Phase 2 results with 95% confidence intervals for all metrics.

Config	Match%	95% CI	Action%	Format%	Time(s)	95% CI	Energy(J)	Mem(MB)
0.5B Q2 K	19.2	[15.8,22.6]	25.5	99.5	3.6	[3.4,3.9]	467	2619
0.5B Q4 K M	18.1	[14.7,21.5]	35.0	96.0	3.7	[3.4,4.0]	472	575
0.5B Q8 0	19.6	[16.1,23.0]	26.0	100.0	5.5	[5.0,6.0]	707	2167

1.5B Q2 K	21.9	[18.3,25.6]	39.0	99.5	6.5	[5.2,7.7]	829	1378
1.5B Q4 K M	26.4	[22.4,30.4]	42.5	99.0	8.1	[7.5,8.6]	1033	1718
1.5B Q8 0	25.7	[21.6,29.7]	40.5	96.5	5.6	[5.2,6.0]	715	2484
3B Q2 K	24.1	[20.3,28.0]	42.5	99.0	9.8	[9.1,10.5]	1253	2439
3B Q4 K M	27.1	[23.0,31.2]	44.5	98.5	9.9	[9.3,10.6]	1274	3309
3B Q8 0	27.0	[22.9,31.1]	44.5	99.0	7.1	[6.6,7.6]	912	5232
7B Q2 K	28.8	[24.7,32.8]	41.5	100.0	12.2	[11.3,13.1]	1566	8176
7B Q4 K M	29.8	[25.8,33.9]	42.5	100.0	16.6	[15.2,18.0]	2128	4950
7B Q8 0	30.3	[26.1,34.5]	43.5	100.0	20.6	[19.2,22.0]	2647	11408

Table 2: Phase 2 complete results — Qwen2.5 across 4 sizes × 3 quantization levels (n=200, 95% CI).

5.2 Size Scaling Analysis

- 0.5B → 1.5B: Avg match improves from 19.0% to 24.7% (+5.7pp, +30% relative). This is the largest single-step gain, indicating 0.5B models lack capacity for structured JSON generation.
- 1.5B → 3B: Avg match improves from 24.7% to 26.1% (+1.4pp, +5.7% relative). Marginal gain per billion parameters drops significantly.
- 3B → 7B: Avg match improves from 26.1% to 29.6% (+3.5pp, +13.4% relative), but at 67% more time and 80% more energy. The Pareto frontier reveals a "knee" at 3B.

5.3 Quantization Impact

- Q8_0 → Q4_K_M: Avg match loss of only 0.2pp across all sizes, while reducing model size by ~50% and memory by ~40%. Q4_K_M is the optimal accuracy-efficiency trade-off.
- Q4_K_M → Q2_K: Avg match loss of 1.6pp, with energy savings of 20-30%. The 7B-Q2_K configuration (28.8% match, 12.2s, 1,566J) outperforms 3B-Q8_0 (27.0%) while consuming only 72% more energy.
- Quantization sensitivity varies by size: 0.5B shows minimal impact (±1.5pp), while 1.5B is most sensitive (up to 5.3pp loss from Q4_K_M to Q2_K).

6. Resource Consumption Analysis

6.1 Memory Footprint

Memory consumption scales primarily with model size and quantization level. Peak RSS ranges from 575 MB (0.5B-Q4_K_M) to 11,408 MB (7B-Q8_0). Table 3 shows the memory footprint across all configurations.

Size	Q2_K (MB)	Q4_K_M (MB)	Q8_0 (MB)
0.5B	2619	575	2167
1.5B	1378	1718	2484
3B	2439	3309	5232
7B	8176	4950	11408

Table 3: Peak memory footprint (RSS in MB) by model size and quantization.

6.2 Energy Efficiency Ranking

Energy consumption is dominated by inference time, as CPU utilization remains at ~95% for all configurations. Table 4 ranks all configurations by energy efficiency (Match% per Joule).

Rank	Config	Match%	Energy(J)	Match/J ×100	Mem(MB)
1	7B Q8 0	30.3	2647	1.145	11408
2	7B Q4 K M	29.8	2128	1.400	4950
3	7B Q2 K	28.8	1566	1.839	8176
4	3B Q2 K	24.1	1253	1.923	2439
5	3B Q4 K M	27.1	1274	2.127	3309
6	1.5B Q4 K M	26.4	1033	2.556	1718
7	1.5B Q2 K	21.9	829	2.640	1378
8	0.5B Q8 0	19.6	707	2.773	2167
9	3B Q8 0	27.0	912	2.961	5232
10	1.5B Q8 0	25.7	715	3.592	2484
11	0.5B Q4 K M	18.1	472	3.831	575
12	0.5B Q2 K	19.2	467	4.110	2619

Table 4: Energy efficiency ranking — sorted by Match%/Joule.

The 1.5B-Q8_0 configuration achieves the highest energy efficiency (0.036 Match/J), followed by 0.5B-Q4_K_M (0.034 Match/J). The 7B models are the least efficient (0.011-0.013 Match/J), consuming ~3× more energy per unit of accuracy compared to 1.5B models.

7. Domain-Level Analysis

We decompose performance across three network slice lifecycle domains (Q4_K_M configurations).

Size	Provisioning Match%/Act%	Scaling Match%/Act%	Conflict Match%/Act%	Overall
0.5B	34.8%/74.3%	14.4%/15.7%	3.1%/11.7%	18.1%/35.0%
1.5B	45.1%/80.0%	25.8%/31.4%	5.4%/11.7%	26.4%/42.5%
3B	52.4%/100.0%	21.6%/20.0%	3.9%/8.3%	27.1%/44.5%
7B	55.4%/100.0%	24.2%/15.7%	6.6%/6.7%	29.8%/42.5%

Table 5: Domain-level performance breakdown (Q4_K_M).

Provisioning intents consistently achieve the highest accuracy (35-42% match, 70-76% action) due to well-defined JSON schemas. Scaling intents show moderate performance (15-22% match) reflecting ambiguity in scaling decisions. Conflict resolution is the most challenging domain (2-4% match), requiring multi-constraint optimization beyond current SLM capabilities.

8. Discussion

8.1 Deployment Recommendations

- Latency-critical edge (<10s): Qwen2.5-3B-Q4_K_M (27.1% match, 9.9s, 1,274J, 3.3GB). Offers 91% of 7B accuracy at 60% cost.
- Accuracy-prioritized: Qwen2.5-7B-Q8_0 (30.3% match, 20.6s, 2,647J). Upper bound for CPU-only inference.
- Ultra-light edge (<2GB): Qwen2.5-1.5B-Q4_K_M (26.4% match, 8.05s, 1,033J, 1.7GB). Competitive accuracy at minimal resource.
- Quantization: Always prefer Q4_K_M over Q8_0. Q2_K acceptable for memory-constrained scenarios but avoid for 1.5B models.

8.2 Limitations

- CPU-only inference: Results are specific to CPU inference. GPU acceleration would change the latency-energy trade-off.
- Energy estimation: Energy is TDP-based approximation (±15%). Direct RAPL measurement was unavailable on VM infrastructure.
- Dataset scope: 200-intent dataset may not capture full production diversity. Future work should validate on operational data.
- Single architecture (Phase 2): Findings are specific to Qwen2.5. Other architectures may exhibit different scaling behavior.
- DeepSeek-R1 exclusion: Reasoning models were incompatible with CPU inference. Future GPU-based evaluation is warranted.

8.3 Future Work

- Multi-agent orchestration with domain-specialized SLMs leveraging observed domain strengths.
- Instruction-tuning on 6G network data, particularly for conflict resolution underperformance.
- GPU benchmarking to establish performance upper bounds.
- Online learning through operator feedback on edge deployments.
- Integration with formal verification for production safety guarantees.

9. Conclusion

This paper presented a comprehensive empirical evaluation of small language models for intent-driven slice lifecycle management in 6G core networks. Through systematic benchmarking of 16 model configurations across 200 networking intents with per-intent resource monitoring, we demonstrated that:

- Qwen2.5-7B achieves the highest intent translation accuracy (29.8% match, 43.5% action) among open-weight SLMs.
- Model size scaling exhibits diminishing returns beyond 3B, with 3B models achieving 91% of 7B accuracy at 60% cost.
- Q4_K_M provides the optimal accuracy-efficiency trade-off (≤1pp loss, ~50% memory reduction).
- Energy consumption is dominated by inference time, making throughput the primary lever for efficiency.
- Domain difficulty varies significantly: provisioning is well-handled even by 0.5B models; conflict resolution challenges all SLMs.
- The 1.5B-3B Q4_K_M configurations represent the sweet spot for edge deployment, balancing accuracy with practical resource budgets.

These findings provide actionable guidance for network operators deploying intent-based automation in 6G infrastructure, and establish a foundation for future research in domain-adapted SLMs for network management.

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